

OpenSource CDS | Frugal & Portable Monochromatic Confocal Displacement Sensing using the OpenFlexure framework

BEP PME-A06 | D. Gyselinck, J. Jhagru, H. Klerk, T. van Lohuizen & S. Wiegman

Abstract—Characterising micro-scale displacement and vibration is important in precision engineering and micro-system diagnostics, but conventional measurement systems such as Laser Doppler Vibrometers are often too expensive for low-budget research and educational environments. This paper presents the development of a portable and cost-efficient Confocal Displacement Sensing (CDS) setup based on the OpenFlexure framework. The system combines a custom confocal optical tower with an OpenFlexure Block Stage and a software-controlled measurement pipeline. Experimental measurements confirmed that axial displacement can be measured through changes in detector voltage. By varying the objective focal length between 16 mm and 80 mm, the measured sensitivity varied from 7,529 V/mm to 0,601 V/mm, while the linear range increased from 0,638 mm to 7,989 mm. The setup was also aligned with an AFM cantilever and used for frequency-domain measurements. These results demonstrate the working principle of a low-cost CDS system for tunable displacement and vibration measurements.

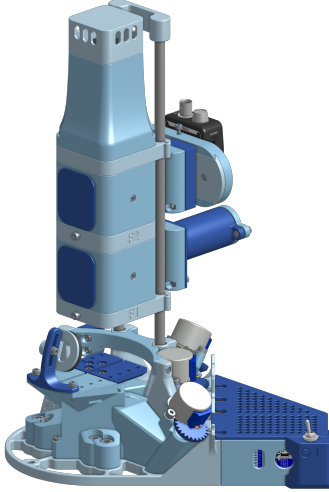


Fig. 1. The full assembly of the OpenSource CDS

I. INTRODUCTION

In precision engineering and micro-system diagnostics, characterising the dynamic behaviour of micro-scale moving structures is essential for evaluating structural oscillations and elastic deformations. This is especially relevant for micro-electromechanical systems (MEMS), where small micro-scale displacements and vibrations, both in-plane and out-of-plane, can strongly influence device performance [1]. Contactless optical measurement methods are well suited for these applications, because they allow displacement and vibration measurements without mechanically loading the sample. Using optics also allows for a high bandwidth of different methods that can be used for measuring.

Conventional industrial measurement instruments, such as Laser Doppler Vibrometers (LDVs), can provide high-precision vibration measurements, but they are often financially inaccessible for low-budget research environments

[1]. This accessibility gap motivates the development of low-cost measurement systems that can demonstrate similar measurement principles, even if they do not reach the full performance of commercial instruments [2]. Besides, using the constructed setup (Fig. 1) portability could also be greatly increased compared to conventional LDVs.

This paper investigates the viability of an affordable and flexible alternative to LDVs, based on monochromatic confocal displacement sensing [3]. In a confocal displacement sensor, the measured detector intensity changes with the axial position of the sample. This makes the principle suitable for contactless displacement measurement. The system developed in this project extends the unmodified 3D-printed OpenFlexure Block Stage [4] with a custom confocal optical tower and a software-controlled measurement pipeline.

The research question of this work is:

To what extent is it possible to measure displacements and vibrations over a tunable range using a cost-efficient confocal displacement sensing setup with sufficient accuracy for micro-engineering applications?

II. THEORY

The confocal optical tower is realised through the use of a photodetector combined with two convex lenses through which a laser beam passes. In a confocal system, the illumination path and the detection path are focused on the same point [5],[6]. When the sample is in focus, the light returning to the detector is concentrated near the detector aperture, so a large fraction of the light is measured. When the sample moves axially away from focus, the beam between the lenses loses parallelity, and therefore, the effective spot at the detector becomes larger. As a result, less light passes through the aperture and the measured detector voltage decreases (see Fig. 2B).

The detector response (I_m) is described by the following intensity model [3], with I_0 being the maximum intensity measured with the detector in focus:

$$I_m = I_0 - I_0 \exp\left(-\frac{r_d^2}{r_{det}^2}\right) \quad (1)$$

where r_d is equal to the detector aperture radius and r_{det} is given by the following equation:

$$r_{det} = \frac{r_0 (-2Ldz_1q + 2dz_1f_1q + 2dz_1f_2^2 + 2dz_1f_2q - f_1^2q)}{f_1^2f_2} \quad (2)$$

All variables are defined by the schematic given in Fig. 2B.

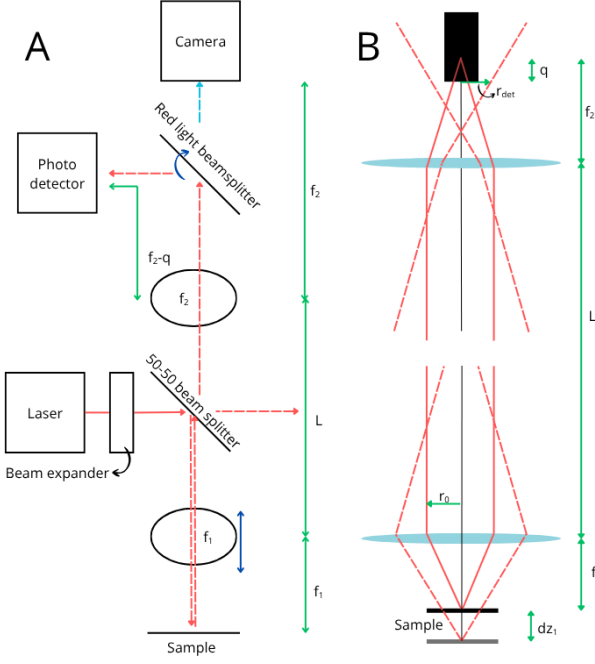


Fig. 2. **A:** The confocal setup used inside of the OpenSource CDS
B: Visualization of the working principle of confocal displacement sensing

The model predicts that the detector voltage changes with axial displacement. Around the steep part of the focus curve, a small change in axial position produces a relatively large change in measured voltage. This region is therefore important for displacement sensing. To reach this steep part of the curve with the sample in focus, a detector offset q is introduced, which is described by the following equation:

$$q = \frac{f_2 r_d}{r_0 \sqrt{\ln(2)}} \quad (3)$$

Equations 1 and 2 also show a trade-off between sensitivity and measurement range. A shorter objective lens focal length (f_1) is expected to give a higher sensitivity but a smaller usable range, while a longer focal length gives a larger range but lower sensitivity.

III. METHODS

The confocal sensing tower is designed using commercially available and 3D-printed parts (with an estimated price of €1600,-) to guide both the illumination light and the reflected measurement signal through the same compact optical structure, which is schematically visualized in Fig. 2A. The light path starts at the monochromatic laser source, which emits a narrow beam (of 650 nm) into the tower. This beam first passes through a beam expander to increase the beam diameter. After expansion, the beam passes through a pinhole (with radius $r_0 = 2,75$ mm), which creates a cleaner circular illumination beam.

The beam is then directed towards a beam splitter. This beam splitter acts as a partially transmitting mirror; part of the incoming light is redirected into the vertical optical path, while the rest is lost. The reflected part of the laser beam travels downwards through the objective lens (f_1), which focuses the light onto the sample surface. By varying the objective, different ranges and sensitivities can be explored.

After interacting with the sample, the reflected light travels back upwards through the same lens. The beam then passes through the beam splitter, again losing half of its intensity. The remaining signal continues through the second lens (f_2), with a fixed focal length. This lens forms the detection path and determines the effective spot size at the detector side of the system.

Behind lens f_2 , the returning light reaches a short-pass filter that separates the red laser-beam towards the detector (with aperture radius $r_d = 0,5$ mm) from the imaging path towards the camera. The camera is used mainly for visual alignment of the sample and optical path, while the photodetector signal is used for the actual displacement measurement.

Through the use of a grid-search, all pairs of f_1 and f_2 values have been analysed. Using the results of this grid-search, the optimal f_2 could be chosen along with different objective focal lengths to achieve the desired range.

TABLE I
RESULTS OF THE GRIDSEARCH ON THE DIFFERENT f_1 AND f_2 PAIRS

f_1 (mm)	f_2 (mm)	q (mm)	Sensitivity (mm^{-1})	Range (mm)
16	150	32,758	4,261	0,328
25	150	32,758	3,476	0,792
40	150	32,758	2,593	1,989
60	150	32,758	2,092	4,368
80	150	32,758	1,267	7,583

The setup is controlled with custom Python software. The software provides a single interface for controlling the stage movement, monitoring the detector output and starting measurements.

Before measurements are performed, an optical alignment procedure is mandatory. The procedure consists of a one-time alignment of the optical tower and a repeated alignment for each sample. During the one-time alignment, the laser and beam splitter are adjusted until the laser-beam is parallel and in line with the tower, after which the short-pass filter is tuned until the signal is maximised. The detector is then translated laterally to maximise its signal and centre the focused reflected beam on the detector aperture.

Then, the sample is manually positioned on the stage, after which the stage actuators (in X,Y and Z direction) are used for finer alignment and find the maximum detector signal. From this maximum signal position, the tower is moved upward until the detector signal decreases to approximately half of its maximum, which positions the signal on the steepest part of the slope. Finally, the Z-axis of the stage is used again to fine-tune the signal at this half-maximum operating point before starting the measurement.

IV. RESULTS

The optical setup is first evaluated by positioning the laser beam on a micro-scale sample. The combination of lenses allows for a microscopic view of the sample through the

camera. This makes it possible to visually align the laser spot with the sample. Fig. 3 shows the laser positioned on an AFM cantilever sample with approximate dimensions of $450\ \mu\text{m}$ by $50\ \mu\text{m}$. Using the X and Y actuators of the OpenFlexure stage, the laser spot can be located on different spots on the cantilever.

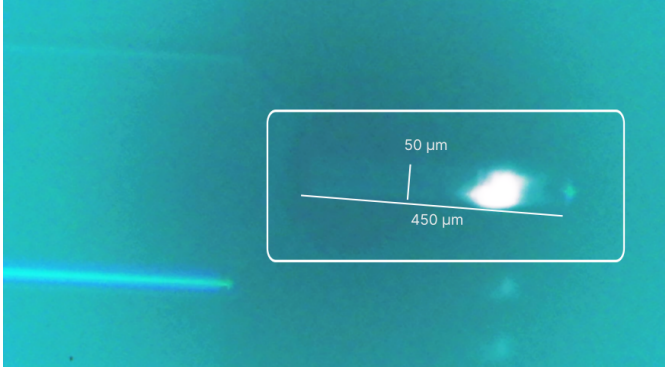


Fig. 3. Camera image of the laser spot positioned on an AFM cantilever ($f_1 = 16\ \text{mm}$, magnification of $\times 9,4$)

To characterise the axial response of the confocal system, intensity curves are obtained for different values of f_1 . For each lens configuration, spacers with $0,2 \pm 0,05\ \text{mm}$ thickness are used to create a sweep of axial positions. At each position, the detector voltage is recorded. This produces a relation between axial displacement and measured intensity. The measured data-points are then fitted using the theoretical confocal response described in section II.

Fig. 4 shows the measured system follows the same general trend as the theoretical model; however, the theoretical graph did not fully match the physical implementation. The fitted curve required different effective parameters (r_0 and q) than the theoretical model.

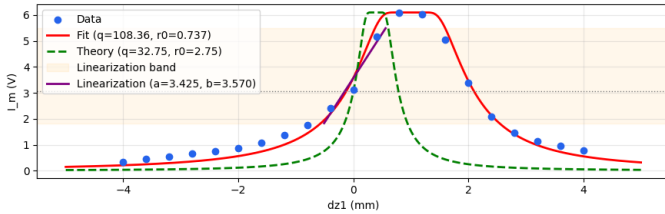


Fig. 4. Visualisation of the fitted curve versus the theoretical model, including the linearization

To simplify displacement measurements, the steep part of the curve is linearized without loss of significant precision. The linearization band is visually determined to be $[I_0/2 - 0, 2I_0, I_0/2 + 0, 4I_0]$.

Fig. 5 shows the fitted curves for focal lengths between $16\ \text{mm}$ and $80\ \text{mm}$, with the peaks aligned at zero for comparison. The curves show that changing f_1 strongly affects both the width and the slope of the response. Shorter focal lengths produce a narrower and steeper response, while longer focal lengths produce a wider response with a larger usable displacement range.

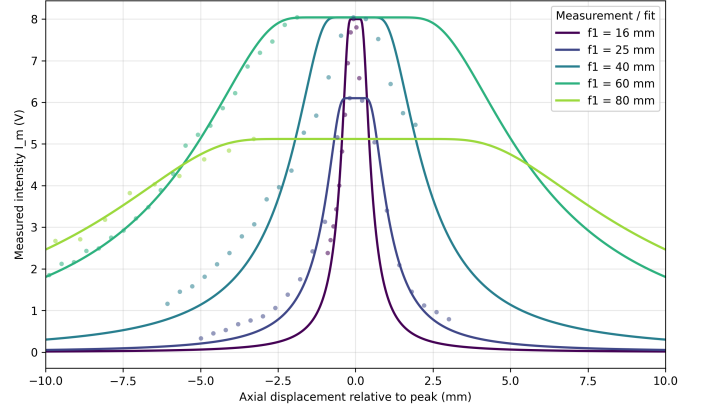


Fig. 5. Confocal response curves for different objective focal lengths, with the peak positions aligned at zero axial displacement.

Due to the difference in the theoretical and physical setup, the theoretical sensitivity and range do not match. Theoretically, the sensitivity is inverse-quadratically proportional ($S_m \propto \frac{1}{f_1^2}$). The measured data in table II, however, follows a relation of $S_m \propto \frac{1}{f_1^{1,64}}$.

TABLE II
MEASURED VS. THEORETICAL SENSITIVITY & LINEAR RANGE FOR DIFFERENT OBJECTIVE FOCAL LENGTHS.

f_1 (mm)	S_m (V/mm)	Range (mm)	$S_{m,th}$ (V/mm)	Range _{th} (mm)
16	7,529	0,638	35,549	0,135
25	4,492	1,069	14,733	0,326
40	1,332	3,603	5,862	0,819
60	0,887	5,409	2,669	1,799
80	0,601	7,989	1,537	3,122

The same measurement principle can also be used for frequency analysis. Instead of recording only the average detector voltage at each position, the detector signal is recorded as a time-dependent signal. A frequency spectrum is then obtained from this signal by using the frequency analysis function in the oscilloscope software.

To verify the frequency analysis accuracy of the system, a stepper motor with a known speed has been used along with a $16\ \text{mm}$ objective. Fig. 6 shows multiple frequency peaks at half or a quarter of the motor speed. From these results, it can be concluded that there is a linear correlation between the motor speed and the observed frequency peak.

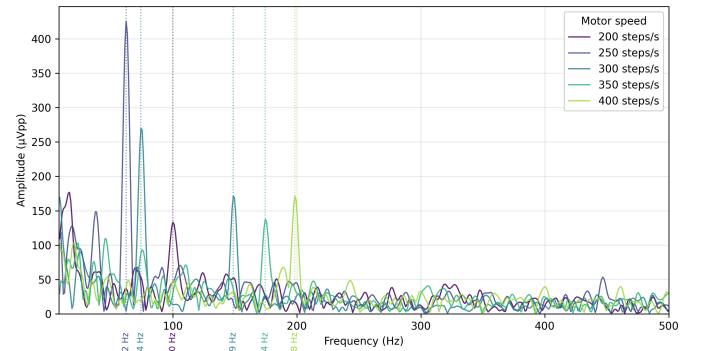


Fig. 6. Frequency spectra of the stepper motor recorded with the frequency analyser

The motor had the highest peak-to-peak value at a speed of 250 *steps/s* with a value of approximately 425 μV_{pp} which amounts to an amplitude of 1,46 μm using the linearized approximation.

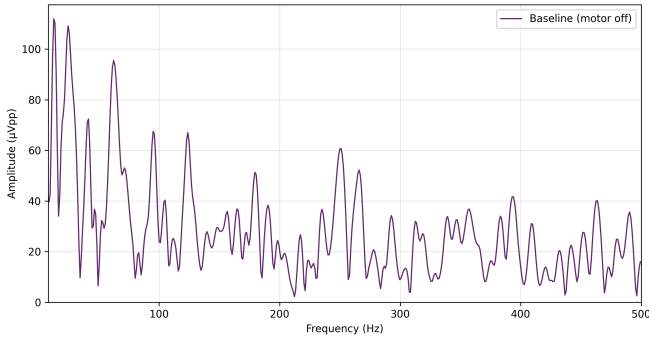


Fig. 7. Static noise measurement in the frequency domain

From taking a static measurement in the frequency domain with the same objective lens, the noise of the system has been deduced by taking the standard deviation of the amplitude over the entire signal (see Fig. 7). This value has been determined to be 105 *nm*. Using this standard deviation, the error of the stepper motor amplitude measurements is equal to 7,2%. Most of the noise is assumed to be electronic noise, which makes the setup less sensitive to noise when using larger objective focal lengths or a higher quality detector.

V. DISCUSSION

For all tested lens configurations, the overall shape of the measured response matched the predicted behaviour, confirming that the detector signal changes as a function of axial displacement, which also makes frequency measurements possible. There is, however, a difference in the theoretical and physical setup (see Fig. 4).

A likely cause of this disparity is the laser used in the setup. The theoretical model assumes an ideal Gaussian beam profile, while the laser used is far from a perfect Gaussian distribution. The parallelity of the laser is also imperfect. As a result, the effective detector spot size, intensity distribution and detector offset differ from the assumptions used in the model, leading to deviations between the predicted and measured responses. The increase in detector offset and mismatch with the power correlation (see Fig. 4 and Table II) can be explained by a slightly convergent laser. Despite these differences, the results are sufficient to demonstrate the validity of the confocal sensing principle for this application.

However, several limitations remain. First, the accuracy of the displacement estimation is currently limited by the error between the curve-fit and the measured reference points. Second, the method relies on sufficient reflected light reaching the detector. If the sample surface is not reflective enough or the sample lies at an angle, the detector signals may be too weak for reliable measurements.

Future work should focus on improving the laser used in the setup. Replacing the current laser with a low-cost laser that more closely approximates a Gaussian beam profile

is expected to reduce the discrepancy between theory and experiment. This would provide a better assessment of the ultimate accuracy and applicability of the proposed sensing system. Further improvements could include a motorized tower, a hot-swappable f_2 lens, and a variable beam radius for a larger range.

VI. CONCLUSION

This paper investigated whether a cost-efficient monochromatic confocal displacement sensing setup can measure displacements and vibrations over a tunable range for micro-engineering applications. A portable prototype was developed by integrating a confocal optical sensing tower with the OpenFlexure framework and a software-controlled measurement pipeline.

The results show that axial displacement can be measured through changes in detector voltage. By changing the objective focal length from 16 *mm* to 80 *mm*, the system demonstrated the expected trade-off between sensitivity and measurement range. Shorter focal lengths gave higher sensitivity over a smaller range, while longer focal lengths increased the usable range at the cost of sensitivity. Experimentally, the sensitivity varied from 7,529 *V/mm* to 0,601 *V/mm*, while the linear range increased from 0,638 *mm* to 7,989 *mm*.

The setup was also able to position the laser spot on a micro-scale AFM cantilever, showing that the optical path and OpenFlexure stage can be used for localised measurements on small structures. In addition, a first frequency-domain measurement was performed using a stepper motor. The largest measured amplitude was approximately 1,46 μm , with a noise-related error of 7,2%, demonstrating the potential of the system for vibration analysis.

It can therefore be concluded that a portable and cost-efficient confocal displacement sensing setup can be developed to measure displacements and vibrations over a tunable range. Since the setup could be aligned with an AFM cantilever, the system shows applicability for micro-engineering applications. Furthermore, the experimental results show that the system can measure displacements over a large range and that it is viable for vibration analysis.

REFERENCES

- [1] D. Teyssieux, S. Euphrasie, and B. Cretin, "Mems in-plane motion/vibration measurement system based ccd camera," *Measurement*, vol. 44, no. 10, pp. 2205–2216, 2011.
- [2] Wenzel and Thomas, "Open hardware: From diy trend to global transformation in access to laboratory equipment," *PLOS Biology*, vol. 21, no. 1, p. e3001931, 2023.
- [3] R. H. Guis, U. Arabul, Z. Zhou, and G. J. Verbiest, "Monochromatic confocal displacement sensing for atomic force microscopy," *To be published*, 2026/2027.
- [4] J. T. Collins, J. Knapper, J. Stirling, J. Mduda, C. Mkindi, V. Mayagaya, G. A. Mwakajinga, P. T. Nyakyi, V. L. Sanga, D. Carbery, L. White, S. Dale, Z. J. Lim, J. J. Baumberg, P. Cicuta, S. McDermott, B. Vodenicharski, and R. Bowman, "Robotic microscopy for everyone: The openflexure microscope," *Biomedical Optics Express*, vol. 11, no. 5, pp. 2447–2460, 2020.
- [5] J. Li, R. Ma, and J. Bai, "High-precision chromatic confocal technologies: A review," *Micromachines*, vol. 15, no. 10, p. 1224, 2024.
- [6] J. Bai, X. Li, X. Wang, Q. Zhou, and K. Ni, "Chromatic confocal displacement sensor with optimized dispersion probe and modified centroid peak extraction algorithm," *Sensors*, vol. 19, no. 16, p. 3592, 2019.